



Probability · Mark Scheme

Statistics Chapter 5: Probability. M = method mark A = accuracy mark (depends on M) B = independent mark ft = follow through

Section A – Venn diagrams

- 1 (a) $0.5 + 0.4 - 0.2 = 0.7$ B1; (b) 0.3 B1.
- 2 (a) $18 + 22 - 30 = 10$ M1 A1; (b) $\frac{8}{30} = \frac{4}{15}$ B1.
- 3 (a) $0.45 + 0.5 - 0.8 = 0.15$ M1 A1; (b) $0.5 - 0.15 = 0.35$ B1.
- 4 (a) Centre 2; $C \cap T$ only 4, $C \cap K$ only 2, $T \cap K$ only 1 B1; C only 12, T only 8, K only 7 B1. (b) $50 - 36 = 14$ B1. (c) $\frac{27}{50}$ B1.
- 5(a) A and C drawn not overlapping B1; $A \cap B = 0.15$, $B \cap C = 0.1$ B1; A only 0.25, B only 0.1, C only 0.2 B1; outside 0.2 B1.
- 5(b) $0.4 + 0.3 = 0.7$ B1.
- 5(c) Regions outside B and C: $0.25 + 0.2$ M1; = 0.45 A1.

Section B – Mutually exclusive and independent events

- 6 (a) 0 B1; (b) 0.75 B1.
- 7 (a) $0.25 \times 0.6 = 0.15$ B1; (b) $0.25 + 0.6 - 0.15 = 0.7$ B1; (c) $0.75 \times 0.4 = 0.3$ B1.
- 8 (a) $P(A)P(B) = 0.4 \times 0.25 = 0.1$ M1; = $P(A \cap B)$, so independent A1. (b) Not mutually exclusive, since $P(A \cap B) \neq 0$ B1.
- 9 (a) $P(S) = 0.3$, $P(S \cap E) = 0.12$ B1 B1. (b) $P(S)P(E) = 0.3 \times 0.6 = 0.18$ M1; $0.18 \neq 0.12$, so not independent A1.
- 10(a) $0.4 + x = 0.7$ M1; $x = 0.3$ A1.
- 10(b) $0.4 + x - 0.4x = 0.7$ M1 A1; $x = 0.5$ A1.
- 10(c) $P(A')P(B) = 0.6 \times 0.5$ M1; = 0.3 A1.

Section C – Tree diagrams

- 11 (a) Correct tree with 0.9, 0.1 on each branch B1; (b) 0.81 B1; (c) $1 - 0.81 = 0.19$ B1.
- 12 $0.4 \times 0.6 + 0.6 \times 0.4$ M1 M1; = 0.48 A1.
- 13 (a) $\frac{4}{10} \times \frac{3}{9} = \frac{2}{15}$ M1 A1; (b) $1 - \frac{2}{15} = \frac{13}{15}$ B1.
- 14 (a) $3 \times 0.7^2 \times 0.3$ M1; = 0.441 A1. (b) $1 - 0.3^3$ M1; = 0.973 A1.
- 15(a) First branches $\frac{5}{12}, \frac{4}{12}, \frac{3}{12}$ B1; second branches out of 11, e.g. after red: $\frac{4}{11}, \frac{4}{11}, \frac{3}{11}$ B1.
- 15(b) $\frac{5 \times 4 + 4 \times 3 + 3 \times 2}{132}$ M1; = $\frac{38}{132} = \frac{19}{66}$ A1.
- 15(c) $1 - \frac{7}{12} \times \frac{6}{11}$ M1; = $\frac{15}{22}$ A1.
- 15(d) Counters are not replaced, so the probability of red changes: the trials are not independent B1.

Section D – Conditional probability

- 16 $\frac{0.12}{0.4}$ M1; = 0.3 A1.
- 17 (a) $0.5 \times 0.4 = 0.2$ B1; (b) $\frac{0.2}{0.5}$ M1; = 0.4 A1.
- 18 Both: $38 + 30 - 44 = 24$ M1 A1; $\frac{24}{38} = \frac{12}{19}$ A1.



19 (a) $0.1 \times 0.9 + 0.9 \times 0.03$ M1; $= 0.117$ A1. (b) $\frac{0.09}{0.117}$ M1; $= 0.769$ A1.

20(a) $0.4 + 0.35 - 0.6 = 0.15$ B1.

20(b) $\frac{0.15}{0.4}$ M1; $= 0.375$ A1.

20(c) $P(A \cap B') = 0.25$, $P(B') = 0.65$ M1; $\frac{0.25}{0.65} = 0.385$ A1.

20(d) $P(A)P(B) = 0.14$ M1; $0.14 \neq 0.15$, so not independent A1.

Roar-level challenge

21 (a) $0.5 \times 0.4 + 0.5 \times 0.2$ M1; $= 0.3$ A1. (b) $\frac{0.2}{0.3}$ M1; $= \frac{2}{3}$ A1. (c) $P(W | L) = \frac{2}{3} \neq 0.5 = P(W)$, so not independent B1.